

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

PRELIMINARY HSC ASSESSMENT TASK

JULY 2008

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- Questions are not of equal value
- Full marks may not be awarded if working is poorly set out or difficult to read.

NAME: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	Total

QUESTION 1

Marks

- a) Factorise $x^3 + 64$ 2
- b) Solve $|3x + 1| \leq 10$ 2
- c) Find integers a and b such that $(3 - \sqrt{2})^2 = a - b\sqrt{2}$ 2
- d) Solve $\frac{x-5}{3} - \frac{x+1}{4} = 5$ 2

QUESTION 2

- a) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$ 2
- b) Find the equation of the line parallel to $3x - y + 4 = 0$ and also having a y intercept of -2 . Express your answer in general form. 3
- c) Find the angle that $2x + y - 3 = 0$ makes with the positive direction of the x -axis. (correct to nearest minute) 2

QUESTION 3

- a) Simplify $(\operatorname{cosec}\theta - 1)(\operatorname{cosec}\theta + 1)$ 1
- b) i) If A is acute and $\sin A = \frac{4}{7}$, find the exact values of $\cos A$ and $\tan A$ 2
- ii) Hence show that $\tan A = \frac{\sin A}{\cos A}$ 1
- c) Prove the identity $\frac{\cos \theta}{1 - \sin \theta} = \sec \theta (1 + \sin \theta)$ 3

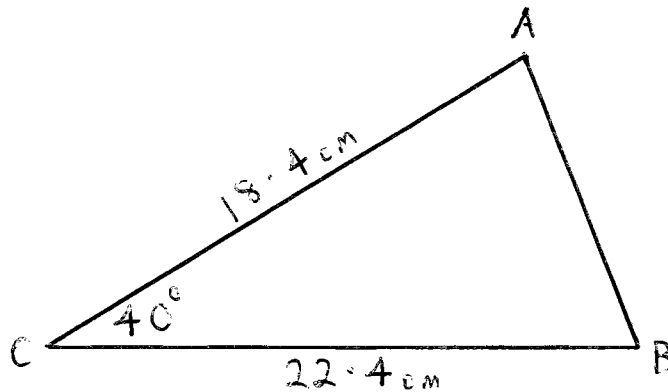
QUESTION 4

Marks

- a) Find the exact value of $\sin(-300)^\circ$ 2
- b) Find the shortest distance between the parallel lines
 $2x - 5y + 10 = 0$ and $2x - 5y - 3 = 0$ 3
(Hint: Find a point on either line).
- c) Solve for $0^\circ \leq x \leq 90^\circ$, 2
 $\sin 50^\circ = \cos(2x - 10)^\circ$

QUESTION 5

a)

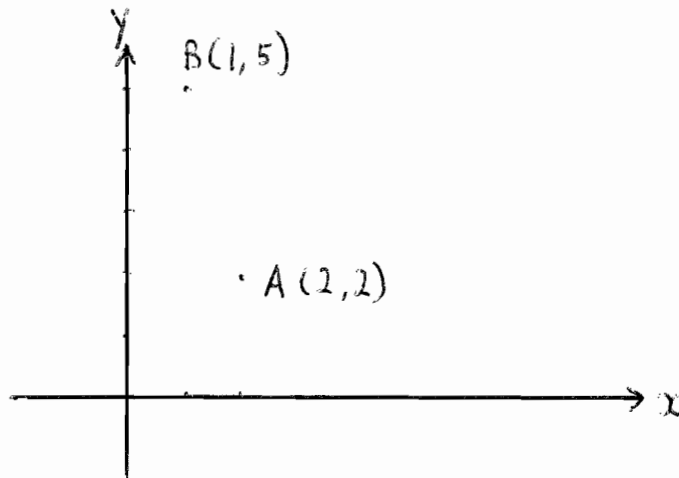


- i) Find the length of side AB correct to one decimal place. 2
- ii) Find the area of triangle ABC correct to the nearest square centimetre 1
- b) Solve $4 \cos \theta = -3$ for $0^\circ \leq \theta \leq 360^\circ$. 2
Give answers correct to the nearest degree.
- c) Sketch the region $2x - y + 3 < 0$ 2

QUESTION 6

Marks

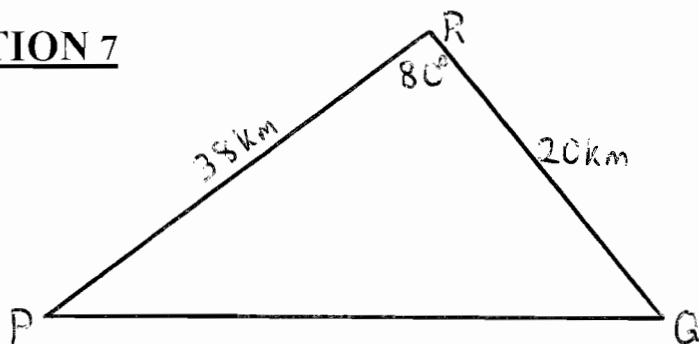
a)



Copy the diagram onto your answer sheet

- | | | |
|------|---|---|
| i) | Find the co-ordinates of M , the midpoint of AB | 1 |
| ii) | Find the gradient of AB | 1 |
| iii) | Show the equation of the perpendicular bisector of AB is
$x - 3y + 9 = 0$ | 2 |
| iv) | Find the co-ordinates of C , which lies on the y - axis and is equidistant from A and B . | 1 |
| v) | The point D lies on the intersection of the line $y = 5$ and the perpendicular bisector $x - 3y + 9 = 0$. Find the co-ordinates of D and mark the position of D on your diagram. | 1 |
| vi) | Find the area of the triangle ABD | 2 |

QUESTION 7



In the diagram, the point Q is due east of P . The bearing of R from Q is 330°T .

- | | | |
|------|--|---|
| i) | What is the size of $\angle PQR$ | 1 |
| ii) | Find the distance PQ (correct to one decimal place) | 2 |
| iii) | What is the bearing of R from P ? (nearest degree) | 2 |

b) Solve simultaneously:

$$x - 2y = 8$$

$$2x + y = 1$$

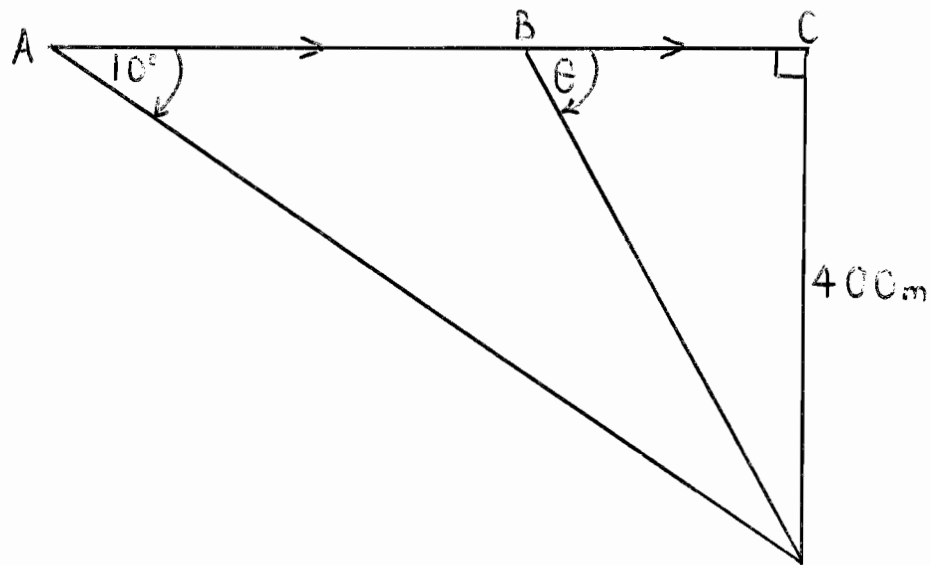
2

QUESTION 8

a) If the points $(-3k, 1)$, $(2, 3)$ and $(k, 4)$ are collinear, find the value of k

2

b)



A helicopter pilot flying horizontally 400m above the ground at a speed of 60km/h, notices a laser gun shone up at him at an angle of depression of 10° . One minute later he has reached point B.

i) Calculate AB

1

ii) Calculate AC (nearest metre)

2

iii) Calculate θ , the angle of depression from B (nearest degree)

1

iv) From B he radios the police and tells them he will hover directly above the offender at C . How long (correct to the nearest second) does it take to fly from B to C ?

2

Teacher's Name:

Student's Name/N°:

July 2008 2 Unit Assessment Solutions

Question 1

$$\begin{aligned}
 a) \quad x^3 + 64 & \\
 &= x^3 + 4^3 \\
 &= (x+4)(x^2 - 4x + 16)
 \end{aligned}$$

(1) (1)

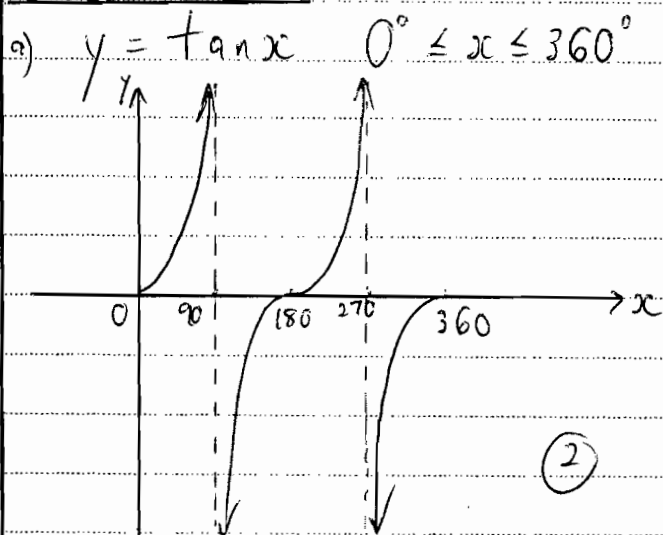
$$\begin{aligned}
 b) \quad |3x+1| &\leq 10 \\
 -10 &\leq 3x+1 \leq 10 \quad (1) \\
 -11 &\leq 3x \leq 9 \\
 -\frac{11}{3} &\leq x \leq 3 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 c) \quad (3-\sqrt{2})^2 &= a - b\sqrt{2} \\
 9 + 2 - 6\sqrt{2} &= a - b\sqrt{2} \\
 11 - 6\sqrt{2} &= a - b\sqrt{2} \\
 a=11, \quad b=6
 \end{aligned}$$

(1) (1)

$$\begin{aligned}
 d) \quad \frac{x-5}{3} - \frac{x+1}{4} &= 5 \\
 4(x-5) - 3(x+1) &= 60 \\
 4x-20 - 3x-3 &= 60 \quad (1) \\
 x-23 &= 60 \\
 x &= 83 \quad (1)
 \end{aligned}$$

Question 2



$$\begin{aligned}
 b) \quad 3x - y + 4 &= 0 \\
 y &= 3x + 4 \\
 m=3, \quad b &= -2 \quad (1) \\
 y &= mx + b \text{ becomes} \\
 y &= 3x - 2 \quad (1) \\
 0 &= 3x - y - 2 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 c) \quad 2x + y - 3 &= 0 \\
 y &= -2x + 3 \\
 m &= -2 \\
 m &= \tan \theta \quad (1) \\
 -2 &= \tan \theta \\
 \theta &= 110.33^\circ \quad (1)
 \end{aligned}$$

Teacher's Name:

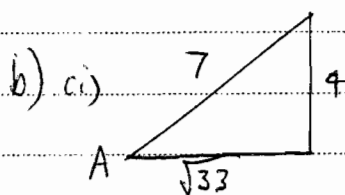
Student's Name/N°:

Question 3

$$a) (\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)$$

$$\operatorname{cosec}^2 \theta - 1$$

$$= \cot^2 \theta \quad (1)$$



$$b) \text{ ci) } \cos A = \frac{\sqrt{33}}{7} \quad (1)$$

$$\tan A = \frac{4}{\sqrt{33}} \quad (1)$$

$$\text{cii) } \tan A = \frac{\sin A}{\cos A}$$

$$= \frac{\frac{4}{7}}{\frac{\sqrt{33}}{7}} \quad (1)$$

$$= \frac{4}{\sqrt{33}}$$

$$c) \frac{\cos \theta}{1 - \sin \theta} = \sec \theta (1 + \sin \theta)$$

$$\frac{\cos \theta}{1 - \sin \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta} = "$$

$$(1) \frac{\cos \theta (1 + \sin \theta)}{1 - \sin^2 \theta} = "$$

$$(1) \frac{\cos \theta (1 + \sin \theta)}{\cos^2 \theta} = "$$

$$\frac{1 + \sin \theta}{\cos \theta} = "$$

$$(1) \sec \theta (1 + \sin \theta) = \sec \theta (1 + \sin \theta)$$

Question 4

$$a) \sin(-300)$$

$$= \sin 60 \quad (1)$$

$$= \frac{\sqrt{3}}{2} \quad (1)$$

$$b) 2x - 5y + 10 = 0$$

(0, 2) lies on above (1)

Now $\perp$ d. from $2x - 5y - 3 = 0$

$$d = \frac{|2 \times 0 + -5 \times 2 - 3|}{\sqrt{2^2 + (-5)^2}} \quad (1)$$

$$d = \frac{13}{\sqrt{29}} \quad (1)$$

$$c) \sin 50 = \cos(2x - 10)$$

$$(1) 50 + 2x - 10 = 90 \quad \text{complementary angles}$$

$$40 + 2x = 90$$

$$2x = 50$$

$$(1) x = 25^\circ$$

Teacher's Name:

Student's Name/N°:

$$\begin{aligned}
 \text{cvi)} \quad A &= \frac{1}{2} b h \\
 &= \frac{1}{2} BC \times 3 \quad (1) \\
 &= 2\frac{1}{2} \times 3 \\
 \underline{A} &= \underline{7\frac{1}{2} \text{ units}^2} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 PQ &= 43.2 / 30.6 \\
 \frac{PQ}{\sin 80} &= \frac{38}{\sin 60} \quad \text{or} \quad \frac{20}{\sin 40}
 \end{aligned}$$

Question 7

$$\begin{aligned}
 \text{a) ci)} \quad &60^\circ \quad (1) \\
 &70^\circ 30'
 \end{aligned}$$

$$\begin{aligned}
 \text{cii)} \quad PQ^2 &= 38^2 + 20^2 - 2 \times 38 \times 20 \times \cos 80 \quad (1) \\
 \underline{PQ} &= \underline{39.7 \text{ km}} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{ciii)} \quad \text{Need } \angle PPQ \\
 \frac{\sin P}{20} &= \frac{\sin 80}{39.7} \quad (1) \\
 P &= 30^\circ
 \end{aligned}$$

$$\therefore \text{Bearing is } \begin{cases} 060^\circ \text{T} \quad (1) \\ 050^\circ \text{T} \\ 063^\circ \text{T} \end{cases}$$

$$\begin{aligned}
 \text{b)} \quad x - 2y &= 8 \quad (1) \\
 2x + y &= 1 \quad (2) \\
 (1) + 2 \times (2)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow 5x &= 10 \\
 \underline{x = 2, y = -3} \\
 (1) \quad (1)
 \end{aligned}$$

Question 8

$$\begin{aligned}
 \text{a)} \quad &(-3k, 1) \quad (2, 3) \quad (k, 4) \\
 &\text{Collinear if}
 \end{aligned}$$

$$\begin{aligned}
 \frac{3-1}{2-(-3k)} &= \frac{4-3}{k-2} \quad (1) \text{ equal gradients} \\
 2k-4 &= 2+3k \\
 \underline{-6 = k} \quad (1)
 \end{aligned}$$

$$\text{b) ci)} \quad \underline{AB = 1 \text{ km}} \quad (1)$$

$$\text{cii)} \quad \tan 10 = \frac{400}{AC} \quad (1)$$

$$\underline{AC = 2269 \text{ m}} \quad (1)$$

$$\begin{aligned}
 \text{ciii)} \quad BC &= AC - AB \\
 &= 1269 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \tan \theta &= \frac{400}{1269} \\
 \underline{\theta = 17^\circ} \quad (1)
 \end{aligned}$$

$$\text{civ)} \quad BC = 1269$$

$$\begin{aligned}
 \text{Time} &= \frac{\text{distance}}{\text{speed}} \\
 &= \frac{1.269}{60} \text{ hours} \quad (1)
 \end{aligned}$$

$$= 76 \text{ seconds} \quad (1)$$

Teacher's Name:

Student's Name/N°:

Question 5

$$a) \text{ i) } AB^2 = 18.4^2 + 22.4^2 - 2 \times 18.4 \times 22.4 \times \cos 40 \quad \textcircled{1}$$

$$AB = 14.5 \text{ cm} \quad \textcircled{1}$$

$$\text{ii) } A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 18.4 \times 22.4 \times \sin 40$$

$$A = 132 \text{ cm}^2 \quad \textcircled{1}$$

$$b) \quad 4 \cos \theta = -3$$

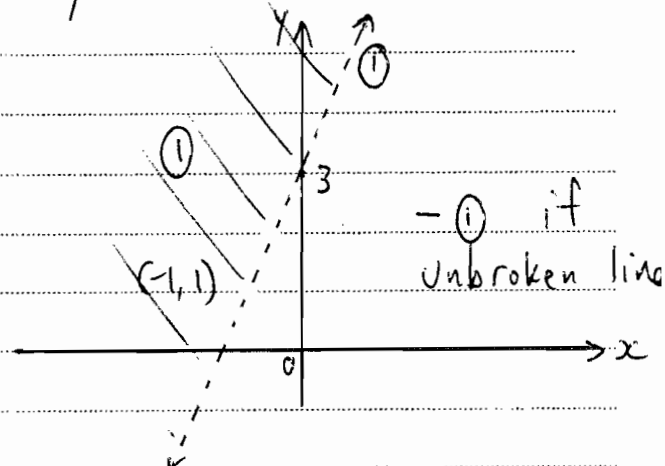
$$\cos \theta = -\frac{3}{4}$$

working $\theta = 41^\circ 25'$

$\sqrt{s}$	A	$\theta = 139^\circ, 221^\circ$
$\sqrt{T}$	c	$\textcircled{1} \quad \textcircled{1}$

$$c) \quad 2x - y + 3 < 0$$

$$y > 2x + 3$$

Question 6

$$a) \text{ i) } M \text{ is } \left(\frac{1+2}{2}, \frac{5+2}{2} \right)$$

$$\left(\frac{3}{2}, \frac{7}{2} \right) \quad \textcircled{1}$$

$$\text{ii) } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - 2}{1 - 2}$$

$$= -3 \quad \textcircled{1}$$

$$\text{iii) } m = \frac{1}{3} \quad \left(\frac{3}{2}, \frac{7}{2} \right)$$

$$y - \frac{7}{2} = \frac{1}{3} \left(x - \frac{3}{2} \right) \quad \textcircled{1}$$

$$6y - 21 = 2 \left(x - \frac{3}{2} \right)$$

$$6y - 21 = 2x - 3$$

$$2x - 6y + 18 = 0 \quad \textcircled{1}$$

$$x - 3y + 9 = 0$$

$$\text{iv) } x = 0 \text{ in } x - 3y + 9 = 0$$

$$y = 3$$

$$\therefore C \text{ is } (0, 3) \quad \textcircled{1}$$

$$\text{v) } x - 3y + 9 = 0 \quad y = 5$$

$$x - 3 \times 5 + 9 = 0$$

$$x = 6$$